

Experience

Two Sigma, Data Clinic

Software Engineer

June 2022 - now

TypeScript, React, Python, FastAPI, Geospatial
(Mapbox, PMTiles, GDAL, GeoPandas, PostGIS),
AWS, LLMs

Data Clinic is the pro-bono initiative of Two Sigma, the hedge fund.

Developed and managed partnership with MobilityData. Led engineering team to launch Mobility Database API. Led junior engineer to launch UI (code here).

Engineering lead on an interactive data viz on flooding + transit (code), presented at SXSW as part of The Census Bureau's Opportunity Project.

Geospatial subject matter expert. Presented a talk titled "Switching to PMTiles" at State of the Map 2023 (and wrote accompanying article).

Developed and managed ongoing partnership with American Red Cross.

Product manager across public good products vertical: led 2025 strategy planning for Scout, a natural language search tool for open data.

Meet and present regularly with nonprofit executive stakeholders.

Princeton University, Electoral Innovation Lab

Senior Software Engineer

January 2020 - June 2022

React, D3.js, Gatsby, Django, Netlify, Mapbox,
tippecanoe, GDAL, GeoPandas, AWS

Engineering and product lead on an interactive visualization, the Redistricting Report Card. Designed and built all site maps, graphics and charts in React, Mapbox, and D3.js. Grew to half a million page views, 50k total users, 1k weekly active users during peak redistricting news cycle (Aug-Dec 2021).

Led data science team on the project. Managed external partner relationship. Project cited by The New York Times, Axios, and local outlets and covered by The Hill.

Also maintained OpenPrecincts. Met regularly with community stakeholders. Gave a talk as part of Mapbox's 2020 election series.

SharedStreets

Software Engineer

October 2018 - October 2019

TypeScript, React/Redux, Node, Mapbox, GeoJSON,
Open source, AWS, CircleCI

Employee #1. Led SharedStreets Road Closures project. Built a geospatial visualization + data editing app and managed pilot w/ PANYNJ.

Gave talks about the project at State of the Map US 2019 and the Transit Techies meetup.

AppNexus

Software Engineer 2 (July 2015 - October
2018)

Led native ads manager web app project (React/Redux/Java) to initial release. Led agile ceremonies for team.

Education

Rutgers University

BS, Computer Science '15

Minor: Political Science

James Dickson Carr Scholar

Yahoo!

Technical Intern

Rutgers Open Systems Solutions

Software Engineer

Saavn

Engineering Intern

Skills

Languages: JavaScript, TypeScript, Python

Frameworks/Libraries: React, Node.js, FastAPI, Django, D3.js, Mapbox/MapLibre

Tools: QGIS, PostGIS, nginx, Figma

Creative: Logic Pro, Adobe Photoshop/Illustrator/InDesign, Final Cut Pro

Projects

Pavement

Founded parking management company, piloted with Asbury Park, NJ

react, leaflet, postgis

Dashability

2nd Place in a 2017 NJ civic engagement hackathon

ES6 (React, Redux), Python (Flask)

Transit feeds ETL - using Dagster to pull GTFS-RT data every minute and save as geoparquet files (2024)

Wifi Heatmap generator - JS tool to make a heatmap of wifi signal strength to figure out where to place additional node in home mesh network (2024)

Instapaper Viewer - building a custom Instapaper scraper + viewer (Node + Svelte) with RAG + LLM features (2024)

NJ Transit stops + inequality - transit + census data d3.js data viz, inspired by the classic New Yorker viz (2024)